



GROW_SCI_A2_W06 · GROW Science teacher note

Staff-facing; review only. Verbatim SoW objective: Consolidate and revisit key ideas through investigation.

Source: GROW scheme of work, Autumn 2 week 6. Pupil goal: plan a fair investigation, record a result and explain a limit.

40-minute sequence and success

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40.

One-minute resets are inside We do 2, Review and Exit. Success: I can name what I change and what I keep the same., I can record what happened in a small test., I can explain one way to make the test more dependable.

Retrieval and I-do check

When	Question	Answer	SoW cell
last lesson	What should match a science claim?	An observation or result from a suitable model.	GROW scheme of work, Autumn 2 week 5
2-4 weeks	Name an effect of added greenhouse gases.	They slow heat loss and add to warming.	GROW scheme of work, Autumn 2 week 4
last term	What do we keep the same in a fair test?	Key conditions apart from the one factor we change.	GROW scheme of work, Autumn 1 week 5

Autumn 1 is the prior half-term. Confirm actual taught coverage before using a retrieval question. Show me: If I change the surface, should I also change the toy? Answer: No. Keep the toy the same. All pupils show by point, card or board.

Words (same as page and organiser):

Word	Tier	Simple definition	Example sentence
variable	Tier 3	a factor that can change in a test	The surface is our changed variable.
fair test	Tier 3	a test that changes one key factor and keeps other key factors the same	We use the same toy in both runs for a fair test.
result	Tier 2	what you observe or measure in a test	Our result is the distance the toy travelled.
repeat	Tier 2	do the test again under the same conditions	We repeat each run to check our result.

Faded example, tiered task and stretch

I do full: "Change rough/smooth surface → keep toy, start and launch the same → measure distance → repeat." We do part-done: "Change → keep toy and the same → measure distance." Removed step: Name surface as the changed factor and a consistent launch as a control. Independent: no filled step. Next fade: Next time give only the question; ask pupils to select the changed factor and two controls without a filled plan.



We do F: Use a visual card for toy, start mark and launch; point to the missing control. M: Name two controls and record a result. H: Explain how a different launch could make a surface comparison misleading. Independent F: Point to the factor that changes. M: Name two controls and record a result. H: Explain why repeat runs help and identify a method limit. Go deeper: Two runs differ. Name a reason and fix a method step. Named harder stretch — Fair test repair: Two repeats differ. Name a safe conclusion, then improve the method for the next test.

The model, source or practical route is: small toy, two safe surfaces, ruler and start marker. Two on-screen interactions are card sort and What should you do if two repeat runs disagree?; native keyboard controls, live feedback, no storage. The diagram is original and simplified, not to scale. Its labels are checked in the facts ledger.

Practical: Predict first: Will the toy go farther on rough or smooth paper? Give a reason. Kit: Small toy, Low launch guide, Rough and smooth paper, Start-line tape, Ruler. Steps: Mark one start and use the same toy; Launch twice on rough paper in the same way; Measure and record each distance; Repeat on smooth paper; change only the surface; Compare runs and name a limit. Results table: Surface | run 1 (cm) | run 2 (cm), Rough | _ | _, Smooth | _ | _. Check against the school risk assessment and CLEAPSS guidance. Keep the track low and clear; avoid raised drops and mains equipment. Do not invent or change results.

Hinge and decision

Question: A team compares carpet and vinyl using one trolley. Which plan lets them test the surface fairly?

Option	Correct?	Diagnosis
Same push; change start and surface.	no	Changes the start as well as the surface, so the comparison is confounded.
Same start; change trolley and surface.	no	Changes the trolley as well as the surface, so the comparison is confounded.
Same surface; change trolley and push.	no	Keeps the surface and changes both the trolley and push; it cannot test surface effect.
Same start and push; change surface.	yes	Uses the new case accurately.

Rule: About 3 in 4 or more right → move on. Otherwise re-model with the named I-do example, then ask the parallel hinge. Parallel: A new team compares felt and foil with a sliding block. What stays the same: block and push, or surface? Answer: block and push; surface changes.

Misconceptions:

Misconception	Spot	Correct
A fair test may change the trolley and surface together.	Pupil chooses B, changing trolley and surface together.	Change only the surface; keep the trolley, start and push alike.
Holding the surface fixed while changing trolley and push tests the surface.	Pupil chooses C, changing trolley and push but not the surface.	The changed surface must be the variable under test.

AQA Entry Level Science style bridge (classroom planning): Plan a controlled comparison and record an observed result. This is a planning bridge, not a claimed unit outcome or code.

Feedback codes: VF · WS · I · NS(+) · E · R · // · ?. At We do 2, record WS or I and actual prompt level; at Review, VF and specific NS(+); at Exit, one self-edit // or read back ?. R means a response to the next step heard by an adult in the next lesson, never auto-awarded. E: evidence captured on the



school's evidence app only if an observable curriculum action occurred, never while distressed. The page stores no response. Pathways: high support with VF/WS; Build: mix of WS/I; Grow: mostly I and self-edit; Launch: independent self-assessment.

Planned adaptations

Broad area	EEF five-a-day strategy	Specific support
Cognition & Learning	scaffolding	Show a three-card plan with two controls already placed, then remove one cue.
Communication & Interaction	cognitive/metacognitive strategies	Let pupils point to the changed factor and say an M/H because sentence before recording.
SEMH	flexible grouping	Offer a predictable kit role and pause before changing stations; re-enter at the last completed step.
Sensory/Physical	technology	Use a low stable track and large ruler marks, or accept a staff-captured photo of a real measurement.
EAL (language, not SEN)	explicit instruction	Pre-teach changed, same and result with gestures; allow home-language planning and M/H test explanation.

Keep M/H challenge open for EAL with the same words and home-language planning. Pupil may point, sign, speak, draw, type, write, use a scribe or have a staff-captured photo. Offer speech-to-text, TTS or immersive reader on the staff device.

Adults, regulation and prompt ladder

Wait and count to 5. → Point to the lesson's diagram, kit or source. → Ask: What is done, and what is the next step? → Model one step, then return the next step to the pupil. → Finish one example together, then fade the help. No physical prompt. Stage scripts: I do "Watch: one thing changes; I keep the other parts the same."; We do "Which control is already on the plan? What is still missing?"; Independent "Point to your result. What did you measure without my help?"

At Independent, one adult works with supported pupils, the other holds the group; choose roles at the start and swap across the week. If only one adult attends, use the actual staffing plan. At Exit, hand off the deepest prompt level and what was independent.

Cold-call / invite bank, answers:

- F: What is our changed factor? → Surface type.; What do we measure? → Distance travelled.
- M: Why keep the same toy? → So toy differences do not affect comparison.; Why use the same launch? → A stronger push could change distance.
- H: What does a repeat help you check? → Whether results are similar under the same conditions.; If repeats disagree, what is a fair response? → Keep the results, check method and repeat; do not alter data.

Regulation: Agree a quiet pause signal and supervised place. Check in before hinge and Independent; point to the last completed step on re-entry. No evidence while a pupil is distressed.

lesson record: GROW_SCI_A2_W06 via staff device or projector only. Gatsby 4: Quality tester: repeat a controlled check before recommending a product change.

Reading gate: GROW 8–11; rough visible-body ARI grade 4.21; age guide 10.21 years; rough FK grade 4.01; 841 pupil-visible words excluding marked adult guidance and navigation. Screening estimate only. This is a rough screen, not an individual assessment.



Pupil-facing risk: A launched toy may hit someone or drop off the track.

Pupil-facing safety: Keep the track low and clear. Retrieve the toy only after it stops.

SLT planning and evaluation

Lesson code: SCI-GROW-A2-W06

SoW reference: GROW scheme of work, Autumn 2 week 6

Exact SoW outcome: Consolidate and revisit key ideas through investigation.

Evidence (scheme of work): Pearson Edexcel Entry Level Certificate in Science 8939 (E1→E3); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Evidence in this lesson: Pearson Edexcel Entry Level Certificate in Science 8939 (E1→E3); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-GROW-A2-W06. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Choose the question card: does surface change how far this toy goes? Point to what we will change. Staff: Offer a predictable seat or standing spot and the first visual cue. No public test.

Starter (3 min): Think back. Say, point or write one answer for each question. Staff: Ask three low-stakes retrieval questions; prior coverage must be checked.

I Do (5 min): I mark a start line. I keep the toy and launch the same. I change the surface, measure distance and repeat. Runs may still vary. Staff: Model the full reasoning step, pre-teach four words and ask the show-me question.

We Do (7 min): The plan has a fixed toy and start line. Add the missing launch control. Name the surface to change. Staff: Show the part-done example, then allow paired talk before an answer. One-minute reset inside this stage.

We Do 2 (5 min): Everyone chooses the fair plan for comparing two surfaces. Staff: All pupils answer the four-option hinge; use the decision rule, not a score.

Independent (8 min): Test two runs per surface or observe. Record distance. Name what changed, stayed the same and could limit the test. Staff: One adult supports agreed pupils, the other holds the group; swap across the week. One-minute reset inside this stage.

Review (4 min): Check the controls and result with a partner. Fix one label before you conclude. Staff: Give a specific verbal comment (VF) and invite one self-edit (//) or read-back (?).

Exit (3 min): Name the change, one control, one result and a next step. Staff: Hear the answer and name how it changes the next example or support. One-minute reset and handoff.

Planned adaptation — cognition learning: Show a three-card plan with two controls already placed, then remove one cue.

Planned adaptation — communication interaction: Let pupils point to the changed factor and say an M/H because sentence before recording.

Planned adaptation — semh: Offer a predictable kit role and pause before changing stations; re-enter at the last completed step.

Planned adaptation — sensory physical: Use a low stable track and large ruler marks, or accept a staff-captured photo of a real measurement.



Planned adaptation — eal: Pre-teach changed, same and result with gestures; allow home-language planning and M/H test explanation.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: cognitive/metacognitive strategies; semh: flexible grouping; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Test two runs per surface or observe. Record distance. Name what changed, stayed the same and could limit the test. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W07. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W07: Review and evidence autumn science (Entry Level Science / UAS).

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Name surface as the changed factor and a consistent launch as a control. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "A team compares carpet and vinyl using one trolley. Which plan lets them test the surface fairly?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Choose the question card: does surface change how far this toy goes? Point to what we will change. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Test two runs per surface or observe. Record distance. Name what changed, stayed the same and could limit the test. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use variable, fair test, result, repeat with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.